

RIVNE INTL, Ukraine

WMO: 333010

Lat: 50.6025N

Lon: 26.1528E

Elev: 758

StdP: 14.30

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.2	4.4	-6.5	4.0	-1.7	-0.3	5.6	4.9	28.8	36.4	26.7	35.3	5.7	110	0.580

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.0	87.3	68.1	84.0	66.7	80.8	65.6	70.8	81.9	69.0	79.5	67.4	77.0	8.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.1	102.5	75.7	65.5	96.7	73.4	64.0	91.8	71.8	35.1	82.2	33.5	79.3	32.2	77.2	77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.5	20.6	18.1	-8.3	91.3	7.3	2.6	-13.6	93.2	-17.8	94.7	-22.0	96.2	-27.3	98.1
			-8.6	73.8	7.1	1.6	-13.7	75.0	-17.8	75.9	-21.8	76.8	-26.9	78.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.9	25.1	27.6	35.7	48.5	58.2	63.9	67.4	66.0	56.8	46.8	37.4	28.6
	DBStd	17.06	10.60	10.47	8.69	7.87	7.36	6.32	5.87	6.02	6.97	7.89	8.61	9.49
	HDD50	3199	773	626	447	118	15	0	0	18	156	384	662	
	HDD65	6904	1238	1046	910	496	233	98	43	61	261	564	827	1127
	CDD50	2080	0	0	2	74	270	416	538	495	221	57	7	0
	CDD65	310	0	0	0	2	22	64	117	91	14	0	0	0
	CDH74	2947	0	0	0	24	250	543	1049	909	169	3	0	0
	CDH80	823	0	0	0	4	45	117	325	297	35	0	0	0
Wind	WSAvg	9.0	10.5	9.9	10.4	9.4	8.5	7.9	7.5	7.2	8.0	8.8	9.7	10.2
Precipitation	PrecAvg	24.1	1.3	1.2	1.3	1.6	2.5	2.9	3.8	2.3	2.3	1.7	1.5	1.6
	PrecMax	32.1	2.8	2.4	3.0	4.6	5.7	5.2	7.4	5.0	6.5	4.5	3.0	3.0
	PrecMin	15.9	0.5	0.6	0.1	0.2	1.0	1.3	0.9	0.2	0.2	0.1	0.2	0.6
	PrecStd	4.1	0.6	0.4	0.7	0.9	1.1	0.9	2.1	1.3	1.5	1.1	0.6	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.4	51.6	62.3	77.2	84.7	86.9	91.1	91.5	84.0	73.5	60.9	48.5
		MCWB	42.9	45.9	50.0	57.0	63.9	68.5	69.9	67.7	63.5	59.5	53.4	45.7
	2%	DB	42.7	45.6	56.1	71.0	80.3	83.5	87.5	87.1	78.8	67.9	55.8	44.9
		MCWB	40.3	42.1	46.6	54.9	62.4	66.8	69.3	67.2	61.7	57.6	51.0	43.2
	5%	DB	39.6	42.4	51.8	66.4	76.6	80.4	83.8	83.2	74.1	62.6	52.1	41.9
		MCWB	37.7	39.8	44.5	52.7	61.4	65.8	68.1	66.3	60.8	55.3	49.3	40.4
	10%	DB	37.0	39.2	47.7	62.4	72.2	76.9	80.2	79.0	69.7	58.6	48.8	39.3
		MCWB	35.5	37.1	42.5	51.0	59.2	64.4	66.8	65.1	59.4	53.1	46.5	37.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.4	46.5	50.9	59.8	67.6	71.2	73.7	72.9	67.0	62.1	54.4	46.5
		MCDB	45.6	50.7	59.6	73.7	78.9	81.1	84.9	85.1	80.2	69.9	58.9	48.4
	2%	WB	40.4	42.4	48.1	56.9	64.7	68.9	71.2	70.3	64.1	58.8	51.9	43.3
		MCDB	42.1	44.9	54.9	67.0	75.5	79.5	82.4	81.4	73.1	65.8	55.1	44.6
	5%	WB	37.8	39.8	45.3	54.5	62.7	67.3	69.4	68.1	62.1	56.2	49.5	40.4
		MCDB	39.3	42.0	50.0	63.5	73.6	77.2	80.3	77.6	70.5	61.6	52.2	41.6
	10%	WB	35.5	37.1	42.7	52.3	60.7	65.6	67.8	66.2	60.1	53.5	46.6	37.9
		MCDB	36.7	38.9	46.8	60.2	69.5	74.3	77.1	76.0	68.1	57.9	48.6	39.0
Mean Daily Temperature Range	5% DB	MDBR	8.3	10.0	13.2	18.0	19.2	18.7	19.0	20.2	17.8	14.8	9.0	7.8
		MCDBR	8.3	12.3	20.8	24.6	25.1	23.3	24.2	26.1	25.7	21.5	14.9	9.8
		MCWBR	7.1	9.4	13.1	11.5	10.8	10.0	9.6	10.0	11.3	12.5	10.5	8.2
	5% WB	MCDBR	7.8	11.8	17.9	21.8	21.8	20.4	21.3	22.1	21.5	19.0	13.6	9.4
		MCWBR	6.9	9.3	12.2	11.1	10.7	10.0	9.6	10.0	11.0	11.7	9.9	8.2
Clear-Sky Solar Irradiance	taub	0.310	0.341	0.378	0.428	0.412	0.406	0.436	0.434	0.400	0.362	0.343	0.307	
	taud	2.221	2.172	2.188	2.134	2.250	2.289	2.219	2.221	2.270	2.358	2.310	2.274	
	Ebn at Noon	220	244	257	257	268	270	259	253	248	237	208	204	
	Edh at Noon	23	31	37	44	41	40	42	40	34	26	22	19	
All-Sky Solar Radiation	RadAvg	269	530	901	1344	1697	1819	1726	1542	1054	625	269	186	
	RadStd	46	82	128	137	156	134	139	102	156	102	28	36	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.95	N/A	N/A	+2.27	N/A	N/A	N/A	-313	+223	+94
(47)	Regional (8 neighbors)	+1.19	N/A	N/A	+1.62	+0.97	+0.79	N/A	-360	+205	+81

Nomenclature: See separate page